

Linear systems and elimination

Positive diagonal preconditioning with a nilpotent stiff limit

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Topic: Diagonal preconditioners for collocation systems

Rating rationale: The problem is a uniform existence assertion for a positive diagonal solution of non-linear spectral equations. It underlies parallel iteration for implicit Runge–Kutta and spectral deferred-correction methods.

Statement

Let $s \geq 2$ be an integer and $0 < c_1 < \dots < c_s \leq 1$ be real collocation nodes. Let

$$\ell_j(t) = \prod_{k \neq j} \frac{t - c_k}{c_j - c_k}, \quad A_{ij} = \int_0^{c_i} \ell_j(t) dt, \quad A \in \mathbb{R}^{s \times s}.$$

Conjecture (van der Houwen–de Swart). For every such node set, does there exist a real positive diagonal matrix $D = \text{diag}(d_1, \dots, d_s)$ such that

$$\rho(I_s - D^{-1}A) = 0?$$

Equivalently, require

$$(I_s - D^{-1}A)^s = 0,$$

or the polynomial identity

$$\det((1-t)D + tA) = \det D \quad \text{for every } t \in \mathbb{R}.$$

No ordering constraint on the positive d_i is imposed. The source states the conjecture for collocation-based Runge–Kutta correctors; the explicit positive-distinct-node quantification here is an editorial formalization of its collocation-matrix setting, also used by the 2025 treatment. The exclusion of a node at zero is essential: such a node gives A a zero row and forces $I_s - D^{-1}A$ to have eigenvalue one. Rescaling positive nodes to $c_s \leq 1$ does not change the existence question.

Numerical significance and known distinctions

For the diagonal-preconditioned iteration of a collocation system on the scalar test equation, the iteration matrix is

$$K(z) = z(I_s - zD)^{-1}(A - D).$$

Its limit as $|z| \rightarrow \infty$ is $I_s - D^{-1}A$. The conjecture thus asks whether the stiff-limit iteration can terminate in at most s steps using a positive diagonal preconditioner, which permits independent stage solves. It does not assert uniform contraction for finite z or absence of transient nonnormal growth.

The case $s = 2$ has an elementary verification. Write $a = c_1 < b = c_2$ and choose

$$d_1 = \frac{a(2b - a)}{2(b - a) + \sqrt{2ab}}, \quad d_2 = \frac{ab}{2d_1}.$$

Both entries are positive, and direct substitution gives $\text{tr}(D^{-1}A) = 2$ and $\det(D^{-1}A) = 1$. Cayley–Hamilton therefore gives $(I_2 - D^{-1}A)^2 = 0$. This verification is included here explicitly; it is not a claim of an all-stage result in the cited papers.

The 2025 MIN-SR-S construction searches for this nilpotence through determinant equations, and explicitly leaves existence at arbitrary stage count unproved. Its theorem on nilpotence of $A - D$ concerns the nonstiff limit, a different matrix. Its variable-sweep MIN-SR-FLEX coefficients also do not supply one fixed D solving the displayed target. Triangular preconditioners can achieve stiff-limit nilpotence by LU splitting, but do not meet the diagonal constraint.

References and status check

- P. J. van der Houwen and J. J. B. de Swart, *Triangularly implicit iteration methods for ODE-IVP solvers*, SIAM Journal on Scientific Computing 18(1) (1997), 41–55. [DOI](#); [CWI PDF](#). §3.2.1, p.46, states the positive-diagonal zero-spectral-radius conjecture; §3.2.2, equation (3.9) and Theorem 3.1, specifies collocation matrices with positive distinct abscissas.
- G. Čaklović, T. Lunet, S. Götschel and D. Ruprecht, *Improving efficiency of parallel across the method spectral deferred corrections*, SIAM Journal on Scientific Computing 47(1) (2025), A430–A453. [DOI and full text](#); [open PDF](#). §2.2, equations (2.11)–(2.13), distinguishes the two limits; §2.2.3, p.A439, Definition 2.10 and equation (2.38), explicitly discusses unproved existence for arbitrary stage counts.

On 2026-09-11, checked both full papers, searched the original title and positive diagonal/nilpotent collocation conjecture, and searched MIN-SR-S existence, proofs and 2026 follow-ups. No theorem or counterexample settling this fixed-diagonal existence target was located. The 2025 paper gives current explicit evidence that the issue remains unresolved; its numerical optimization is not an all-node, all-stage existence proof. This is a bounded check. The target is distinct from the [LU-based spectral disk question](#) and from existing incomplete-factorization entries.

Reviewed partial results — 13 September 2026

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The [submitted report](#) proves existence for all two- and three-node sets (Section 2), scaled Laguerre node sets at every stage count (Theorem 3.2), and sufficiently clustered positive nodes at each fixed stage count (Theorem 5.3). Section 7 supplies six exact point certificates and six node-neighborhood certificates in dimensions four through seven. The neighborhood sizes depend on the stage count; these results do not cover arbitrary prescribed nodes in every dimension.

The eleven-stage obstruction in Theorem 6.2 excludes a stronger nonnegative-coefficient eigenpolynomial ansatz, not positive diagonals in general. The [structural supplement](#), Sections 2–4, gives a diagonal-similarity reduction and Vandermonde principal-minor identity, plus a non-collocation auxiliary counterexample. The two overlapping bundles are recorded as one submission.

These restricted results passed an [independent informal AI-agent review](#). **Partially resolved:** existence for arbitrary prescribed positive distinct nodes at every stage count $s \geq 4$ remains open outside the established families.

[Submission record and verified affiliation](#). Substantial AI assistance is disclosed. The review is informal automated review, not external human peer review or formal verification. No Lean verification was performed.